STA 643 Project 2: Turbine Blade Optimization

This is a comprehensive project on the design and analysis of computer experiments, which will be worth 20% of your final grade. The project will consist of two parts: a research report (15%) and a short presentation (5%).

Problem Description: You are a data scientist working on an interdisciplinary collaboration with aerospace engineers on designing turbine blades for gas turbine engines, which are widely used as power generation devices for aircraft and marine propulsion. Turbine blades are crucial for the operation of such engines; they extract the high-temperature and high-pressure gas from combustion, and release this in a lower-temperature and lower-pressure environment to create propulsion. The *design* of turbine blades is a highly challenging task. In modern gas turbines, the inlet gas temperature can be as high as 1500°C, which exceeds the melting temperature of most metal airfoils. This harsh operating condition leads to stress and deformation of the turbine blades, which leads to quicker degradation of the engine. Given a limited experimental budget, the project aims to find a *material design* and *cooling schedule* for the turbine blades to minimize blade stress and deformation.

Your aerospace collaborators provide you with the following information:

- Since physical experiments are expensive to prototype and run, computer experiments are used instead to reduce costs. The simulator makes use of a *finite-element model (FEM)* solver, which simulates the physical deformation of the blade during operation. There are six *input variables* for this simulator:
  - *Young's Modulus* x_1: This is a material property of blade metal, which quantifies how easily it can stretch and deform under stress. The larger this is, the less the material stretches under stress. The desired design range is $[200 \times 10^9, 300 \times 10^9]$ Pa.
  - *Poisson's Ratio* x_2: This measures the magnitude of Poisson's effect for the blade metal, the phenomenon where the material tends to expand in directions perpendicular to the direction of compression. The desired design range is $[0.1, 0.49]$ (unitless).
  - *Coefficient of Thermal Expansion* x_3 (CTE): This measures the rate at which the blade metal material expands with an increase in temperature. The desired design range is $[5 \times 10^{-6}, 1.5 \times 10^{-5}]$ K^{-1}.
  - *Thermal Conductivity* x_4: This measures the ability of the blade metal material to conduct heat. Gases generally have low thermal conductivity, and pure metals have high thermal conductivity. The desired design range is $[5, 15]$ W/m/K.

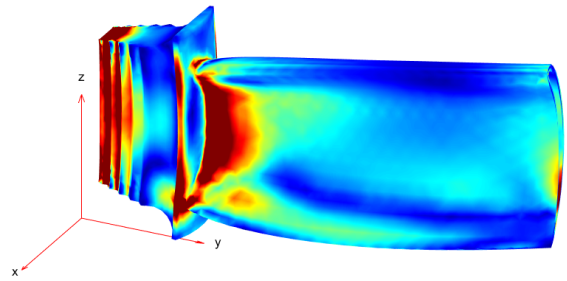

Figure 1: *The simulated von Mises stress profile for a select turbine blade design.*

- *Internal Cooling Air Temperature* x_5: The desired internal cooling temperature for turbine blades, such that the blade can withstand extreme temperatures. The desired design range is $[50, 350]$ °C.
- *Pressure Load on Suction* x_6: The pressure load on the suction end of the turbine. A pressure drop is needed from the pressure and suction ends of the turbine to facilitate movement of high-pressure gas. The desired design range is $[1.0 \times 10^5, 4.8 \times 10^5]$ Pa.

- The simulator returns *two scalar outputs*: the maximum stress over the turbine blade, and its maximum displacement (which measures the amount of deformation). Figure 1 visualizes the von Mises stress profile from which these quantities are computed.

- The primary goal of the project is to identify an optimal specification of blade material properties and cooling conditions that *minimize maximum stress* over the turbine blade. But there are two important additional considerations:

  - *Displacement Constraint*: For this gas turbine system, it is known that a maximum displacement greater than $d^* = 1.3 \times 10^{-3}$ will result in *immediate* failure of the engine. Your collaborators thus wish to consider only blade designs with maximum displacement less than d^*.
  - *Operational Uncertainties*: In real-life implementation, it is difficult to precisely control the operational inputs x_5 and x_6, the internal cooling temperature and pressure load within the turbine. Current technologies allow these inputs to be controlled within tolerance levels of ± 2°C and $\pm 1.0 \times 10^4$ Pa, respectively. Your collaborators wish to ensure that your optimized turbine blade design performs well given such operational uncertainties, both in terms of minimizing maximum stress and satisfying the required displacement constraint.

  These considerations should be incorporated in some fashion into your modeling and/or sequential design procedure.

- Given a computational budget, your analysis is limited to *a maximum of 300 computer experiment runs*. You should carefully consider how to allocate such runs for initial design and sequential design. You should allocate some runs for validating your optimized design. A small amount of bonus points will be given to the group with the best design given the displacement constraint and operating uncertainties, which we will simulate in class.

- You can perform one simulation run by running the following code in MATLAB (make sure you have `simulator.p` in your current directory):

```
>> tic
ymod = 227E9; %Young's modulus
prat = 0.27; %Poisson's ratio
cte = 12.7E-6; %Coefficient of thermal expansion
therm = 11.5; %Thermal conductivity
ctemp = 150; %Internal cooling temperature
press = 4.5E5; %Suction pressure load
[stress, displ] = simulator(ymod,prat,cte,therm,ctemp,press);
toc
```

  This takes around 30 seconds on my laptop, but in reality, such experiments can be much more time consuming!

Other project considerations:

- *Groups*: You may choose to work individually or in a group. You are free to form your own groups, but keep in mind that an ideal group should cover a diverse range of skills in modeling, coding, writing and presentation. Also keep in mind that we will be expecting a more comprehensive analysis from larger groups.
- You are more than welcome to use new methods beyond what was taught in class. There are quite a few complexities in this problem which were not discussed, and you are encouraged to learn and apply new methodology to address these difficulties.
- *Tips*:
  - *Start early*! Meaningful design and analysis require careful thought, and is difficult to do well when crammed in a couple of days. Starting early will give you ample opportunities to seek feedback and improve your project.
  - *Get feedback*! You are strongly encouraged to proactively seek feedback from me and the TAs during *all* stages of your project. Seeking and implementing feedback will improve the quality of your project, which may result in better grades.

Deliverables: Each group should submit **a research report** (no more than 10 pages long, excluding appendices) detailing your methodology, analysis and recommendations for the jet turbine blade design. Each group will also make a brief **10 minute presentation** on your findings. **The report will be due Apr 6th, 11:59pm on Canvas, and the presentation will be held in-class Apr 7th.** Your target audience consists of your scientific collaborators who have working knowledge of class material, but is interested in novel methods and applications for the target problem - you should "sell" your methodology to them, and showcase your value as a collaborator.